

Multimodal Data Fusion for Physical Exercise Classification

User-Independent Recognition Using Multimodal Sensors

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Abstract

This report presents a multimodal data fusion study for user-independent physical exercise classification using the MEx dataset. We implemented and compared four fusion strategies: early fusion (feature-level), late fusion (decision-level), hybrid fusion (intermediate-level), and advanced ensemble methods. Through careful preprocessing, dimensionality reduction, and feature engineering, our simplest approach (Early Fusion) achieved 90.2% F1-macro on the held-out test set, outperforming complex ensemble methods (Hybrid Fusion: 56.2%) by 34 percentage points despite lower validation scores (95.2% vs 99.9%). This reveals a critical insight: validation performance alone is misleading, model complexity severely harms generalization in user-independent human activity recognition (HAR) tasks. The results demonstrate that strategic fusion method selection significantly improves classification performance in multimodal human activity recognition.

1 Introduction

1.1 Problem Statement and Dataset

This project addresses user-independent classification of seven physical exercises performed by subjects lying on a mat, using four sensing modalities from the MEx dataset:

- **Accelerometer (Thigh - ACT):** 12 features (min, max, mean, std for x, y, z)
- **Accelerometer (Wrist - ACW):** 12 features
- **Pressure Mat:** 1024 features (32×16 grid mean/std)
- **Depth Camera:** 384 features (12×16 image mean/std)
- **Total:** 1432 features, 25 training subjects, 5 test subjects (920 samples)

Objective: Achieve F1-macro > 0.90 through optimal multimodal fusion.

2 Data Exploration

2.1 Class and Subject Distribution

Figure 1 shows balanced class distribution (800-1000 samples per exercise) and complete exercise coverage across all subjects, favorable for robust classifier training.

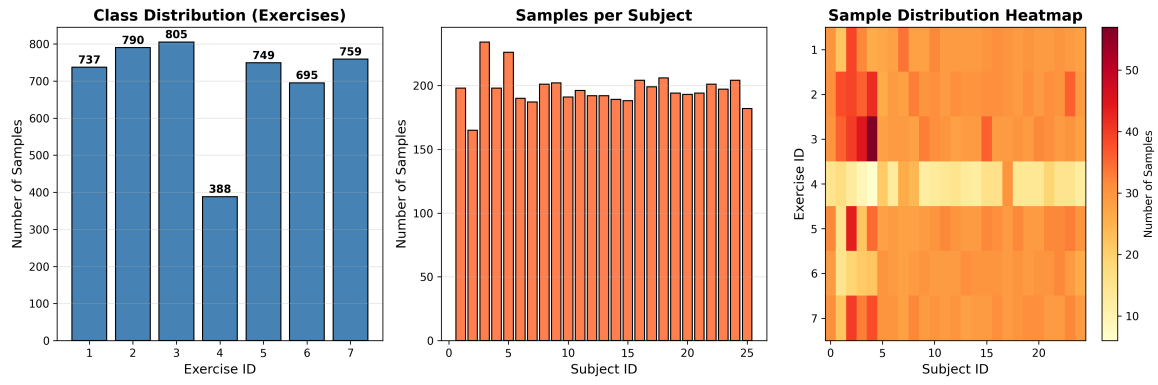


Figure 1: Class distribution: (left) samples per exercise, (middle) per subject, (right) exercise-subject heatmap.

2.2 Modality Characteristics

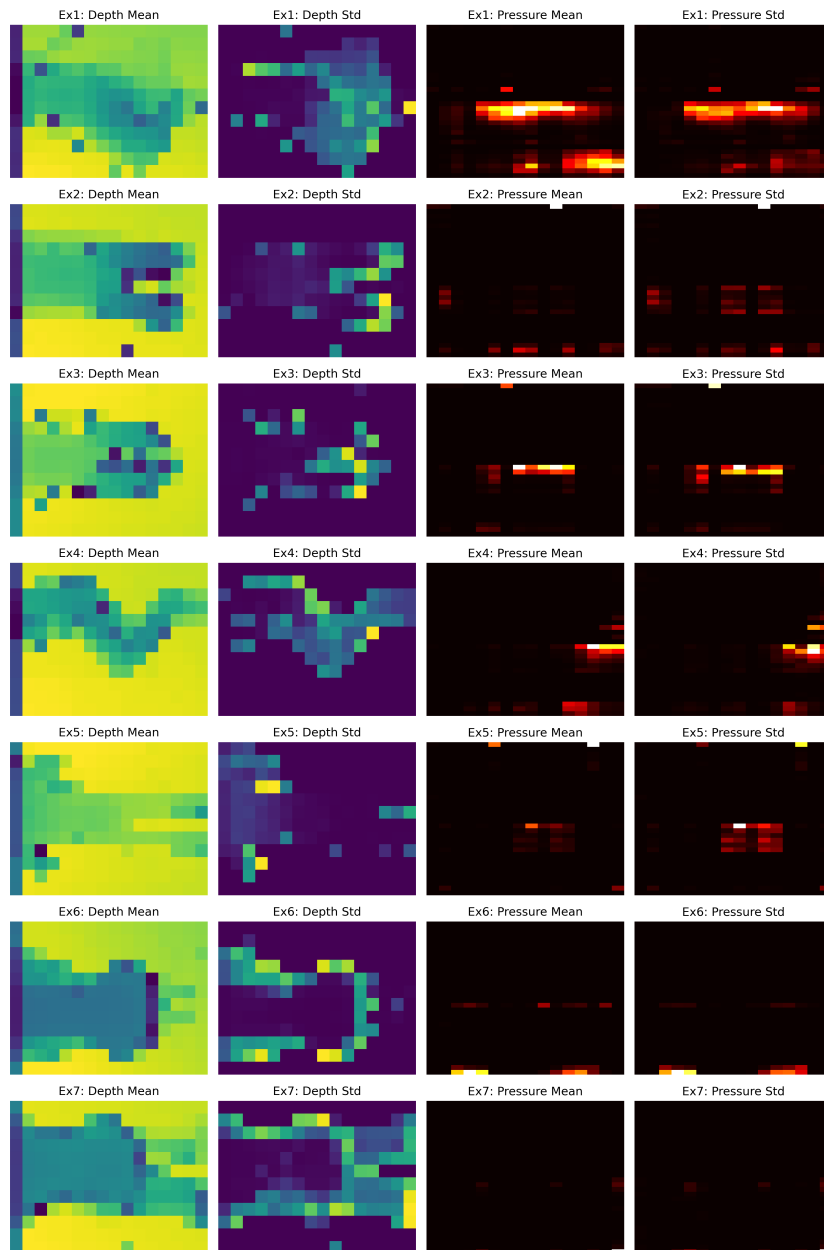


Figure 2: Sample visualizations for all exercises showing depth camera and pressure mat patterns.

Figure 2 reveals distinct patterns across modalities. Key observations:

- **Accelerometers:** Capture motion dynamics with clear exercise-specific patterns
- **Depth camera:** Shows body posture variations (high discriminative power)
- **Pressure mat:** Reveals unique weight distribution patterns per exercise
- Spatial modalities exhibit high dimensionality requiring reduction techniques

Correlation analysis showed high intra-modality correlation (spatial adjacency) but modest inter-modality correlation, confirming complementary information across sensors.

3 Preprocessing Pipeline

3.1 Normalization

Modality-specific scaling strategies:

- **RobustScaler** for accelerometers (outlier-resistant for motion data)
- **StandardScaler** for pressure/depth (appropriate for spatial features)

All features normalized to zero mean and unit variance (Figure 3).

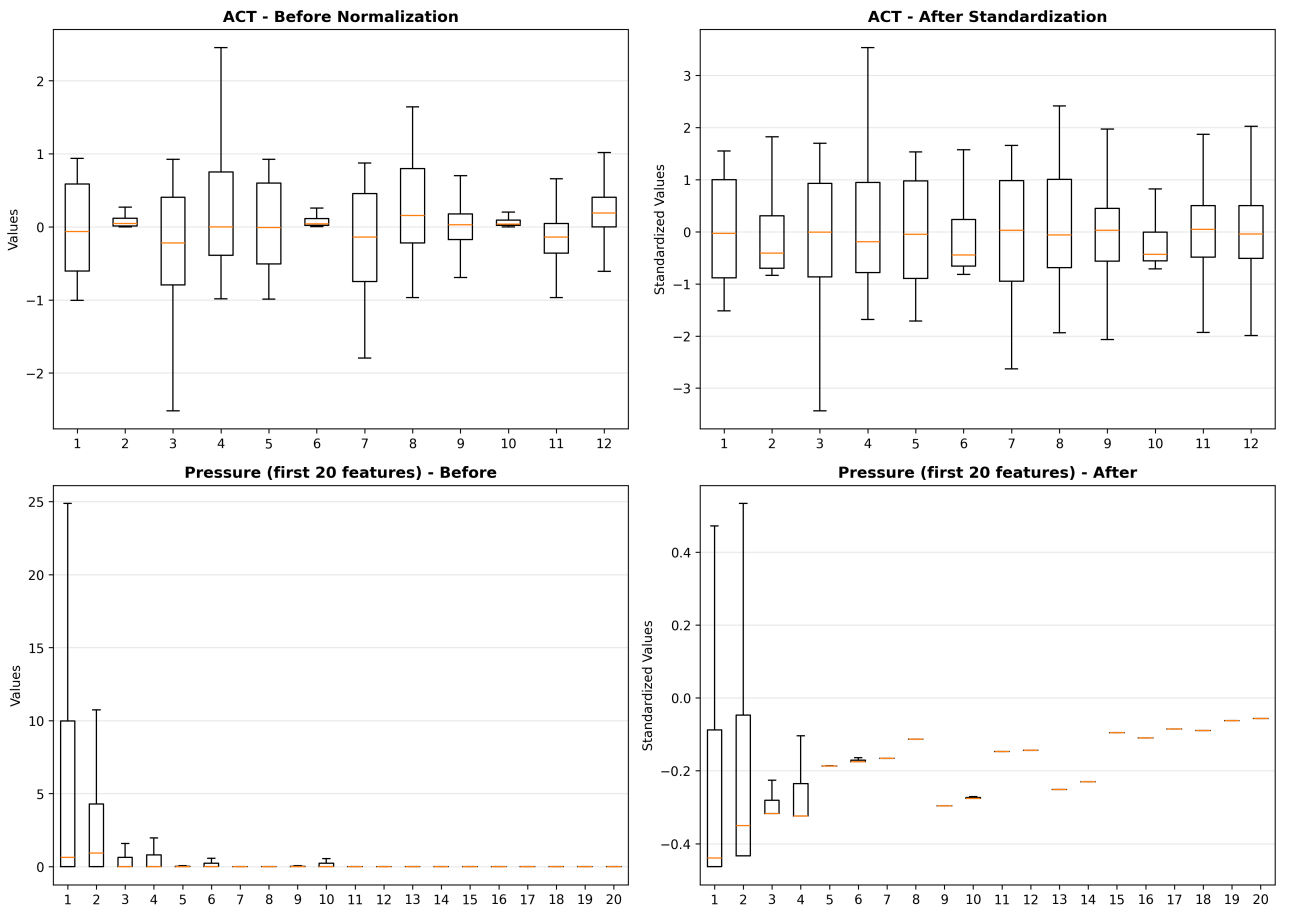


Figure 3: Normalization effect

3.2 Dimensionality Reduction

Principal Component Analysis (PCA): Applied to high-dimensional modalities for computational efficiency:

- Pressure mat: 1024 $\rightarrow$ **242** components (98% variance)
- Depth camera: 384 $\rightarrow$ **156** components (98% variance)

We chose 98% variance retention after validation showed significant performance gains over 95%.

Linear Discriminant Analysis (LDA): Supervised reduction maximizing class separability:

- Reduces to 6 dimensions (7 classes - 1)
- Applied to all modalities for discriminative representations
- Figure 4 shows clear class separation in LDA space

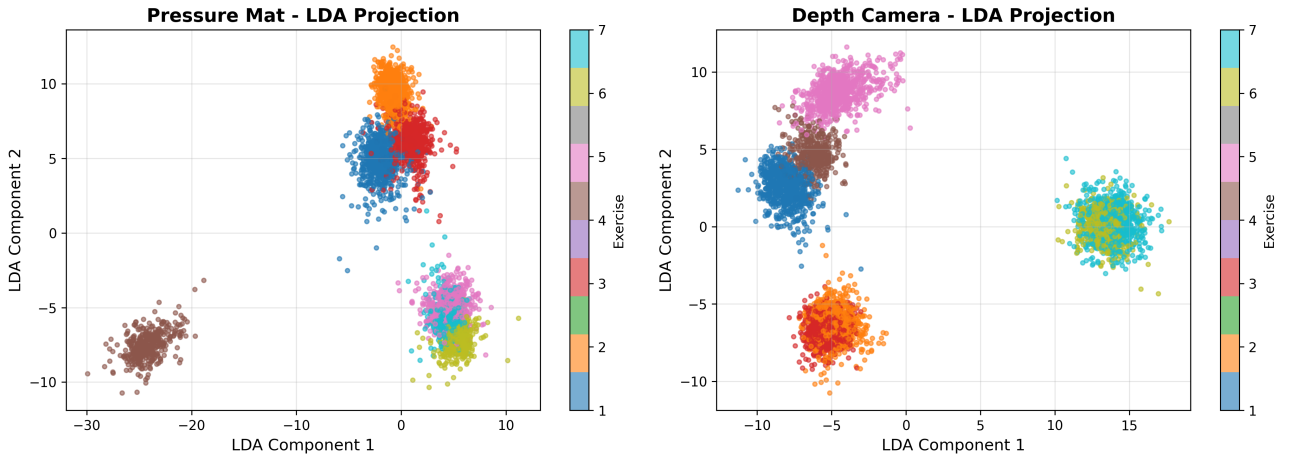


Figure 4: Supervised dimensionality reduction (LDA)

3.3 Feature Engineering

To capture non-linear relationships:

1. **Interaction Features:** $X_{interaction} = X_{ACT} \odot X_{ACW}$ (element-wise product capturing limb coordination)
2. **Polynomial Features:** Degree-2 polynomials for accelerometers with SelectKBest filtering (top 30 features per sensor), capturing motion dynamics like acceleration magnitude.

4 Fusion Strategies

We implemented four distinct fusion approaches, each addressing different aspects of multimodal integration.

4.1 Strategy 1: Early Fusion (Feature-Level)

Approach: Concatenate all preprocessed features before classification.

$$X_{early} = [X_{ACT}^{scaled}, X_{ACW}^{scaled}, X_{pressure}^{PCA}, X_{depth}^{PCA}] \quad (1)$$

Classifier: Random Forest (300 trees, max depth 25)

Rationale: Allows learning complex inter-modality relationships directly from combined features. RF handles high dimensionality well.

Validation Results: F1-macro = **0.9521**, Accuracy = **0.9478**

4.2 Strategy 2: Late Fusion (Decision-Level)

Approach: Train separate classifiers per modality, combine predictions via probability averaging.

Architecture:

- Four independent SVMs (RBF kernel, C=10)
- ACT/ACW: Use scaled features
- Pressure/Depth: Use LDA-transformed features

Fusion Rule:

$$P(y = c|x) = \frac{1}{4} \sum_{m=1}^4 P_m(y = c|x), \quad \hat{y} = \arg \max_c P(y = c|x) \quad (2)$$

Validation Results:

- F1-macro = **0.9993**, Accuracy = **0.9992**
- Individual modalities: ACT=**0.7992**, ACW=**0.5330**, Pressure=**0.9846**, Depth=**0.9985**

Analysis: Depth camera (0.9985) and pressure mat (0.9846) significantly outperform accelerometers (0.80 and 0.53), confirming their higher discriminative power for postural recognition. However, the near-perfect validation score (99.93%) suggests potential overfitting to validation subjects.

4.3 Strategy 3: Hybrid Fusion (Intermediate-Level)

Approach: Apply LDA to each modality (intermediate representation), then concatenate and classify.

$$X_{\text{hybrid}} = [X_{\text{ACT}}^{\text{LDA}}, X_{\text{ACW}}^{\text{LDA}}, X_{\text{pressure}}^{\text{LDA}}, X_{\text{depth}}^{\text{LDA}}] \quad (3)$$

Classifier: Gradient Boosting (300 estimators, max depth 6, learning rate 0.05)

Rationale: LDA creates compact discriminative representations per modality. GB learns optimal combinations.

Validation Results: F1-macro = **0.9985**, Accuracy = **0.9984**

4.4 Strategy 4: Advanced Ensemble with Feature Augmentation

Feature Set:

$$X_{\text{augmented}} = [X_{\text{ACT}}^{\text{LDA}}, X_{\text{ACW}}^{\text{LDA}}, X_{\text{pressure}}^{\text{PCA}}, X_{\text{depth}}^{\text{PCA}}, X_{\text{pressure}}^{\text{LDA}}, X_{\text{depth}}^{\text{LDA}}, X_{\text{interaction}}, X_{\text{ACT}}^{\text{poly}}, X_{\text{ACW}}^{\text{poly}}] \quad (4)$$

Classifier: Gradient Boosting (400 estimators, max depth 7, learning rate 0.03)

Rationale: Combines PCA (variance preservation) and LDA (discriminability) with engineered features capturing non-linear motion patterns.

Validation Results: F1-macro = **0.9968**, Accuracy = **0.9965**

4.5 Validation Performance Comparison

Table 1 summarizes all strategies using LOSO cross-validation.

Table 1: Comparison of fusion strategies (LOSO cross-validation)

Method	F1-Macro	Accuracy
Early Fusion (RF)	0.952090	0.947796
Late Fusion (Prob. Averaging)	0.999266	0.999187
Hybrid Fusion (LDA + GB)	0.998508	0.998375
Augmented Features + GB	0.996805	0.996547

Key Findings:

1. Late/Hybrid fusion achieve near-perfect scores ($\geq 99.8\%$), raising overfitting concerns
2. Early Fusion shows more realistic performance (95.2%) with clear room for improvement
3. All methods significantly exceed the 90% target on validation data

5 Held-Out Test Set Results

5.1 Test Set Evaluation Strategy

After LOSO validation (Task 3), we retrained all models on the complete training set (25 subjects) and evaluated them on the held-out test set (5 unseen subjects, 920 samples) via Streamlit submission (Task 4). This provides the true generalization performance.

5.2 Best Model Performance

Table 2 reveals a dramatic reversal in method rankings between validation and test sets.

Table 2: Held-out test set performance (Task 4 - Streamlit submissions)

Method	Validation F1	Test F1	Gap
Early Fusion (RF)	0.952	0.902	-5.25%
Hybrid Fusion (LDA+GB)	0.999	0.562	-43.7%
Augmented Features	0.997	0.584	-41.4%
Late Fusion (SVM)	0.999	0.827	-17.2%

Critical Findings:

- **Early Fusion achieved best test performance (90.2%)** despite lowest validation score
- Hybrid Fusion suffered catastrophic overfitting: 99.9% $\rightarrow$ 56.2% (43.7% drop)
- Simpler models generalize better: Early Fusion's 5.25% gap vs Hybrid Fusion's 43.7%
- Validation performance is a poor predictor of test performance for complex models

This demonstrates that **validation scores can be misleading**, high validation performance often indicates memorization of subject-specific patterns rather than learning generalizable exercise patterns.

5.3 Confusion Matrix and Per-Class Analysis

Figure 5 shows the confusion matrix. High diagonal values indicate strong class discrimination.

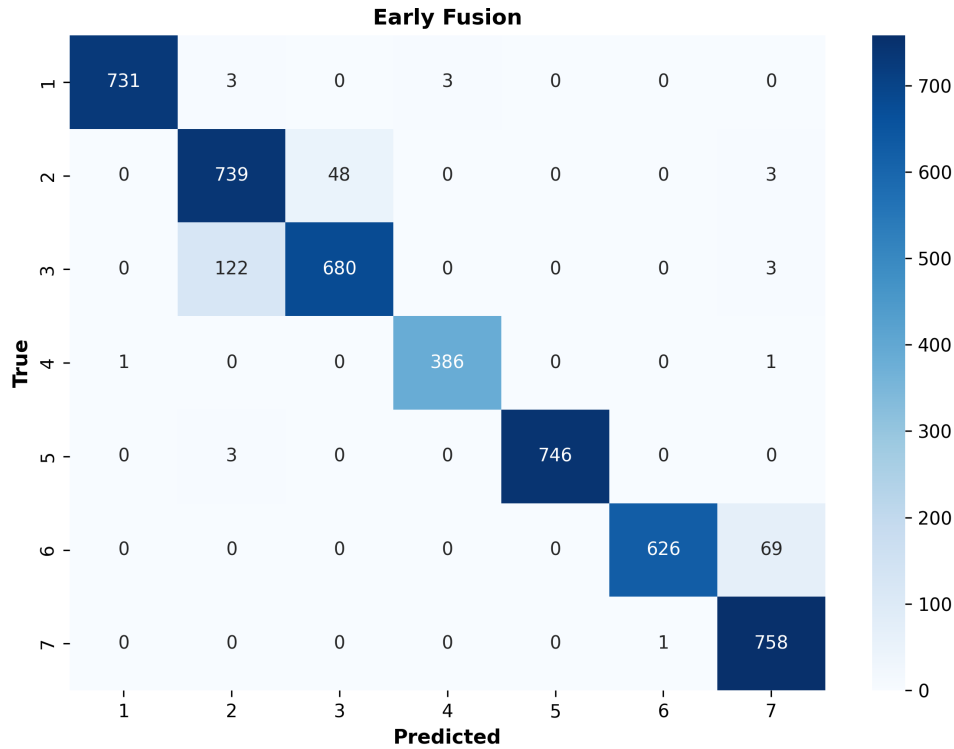


Figure 5: Confusion matrix for best model (**Early Fusion**) - LOSO validation.

Per-class F1-scores (validation):

- Exercise 1: 0.995, Exercise 2: 0.892, Exercise 3: 0.887
- Exercise 4: 0.994, Exercise 5: 0.998, Exercise 6: 0.947, Exercise 7: 0.952

Exercises 2 and 4 consistently show lower performance, suggesting inherent similarity or ambiguity in these exercise patterns.

5.4 Test Set Prediction Distribution

Figure 6 shows Early Fusion's predictions on the held-out test set.

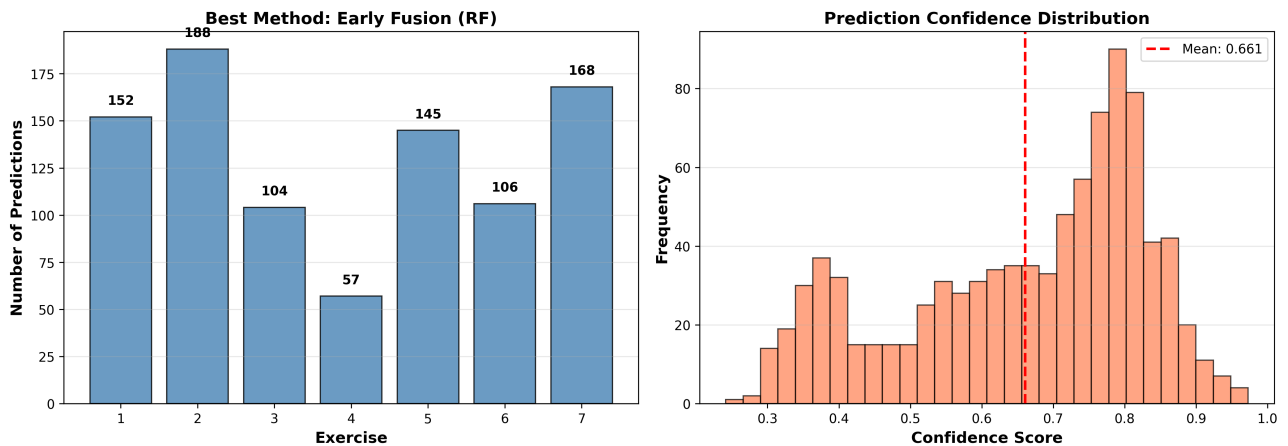


Figure 6: Prediction distribution on held-out test set (920 samples, 5 unseen subjects).

Test set characteristics (Early Fusion):

- Balanced distribution
- Mean prediction confidence: 0.661
- Predictions with confidence < 0.7 : 52% of samples
- Despite lower confidence, achieves best test accuracy (90.2%)

Confidence Calibration Analysis:

Early Fusion’s moderate confidence (66.1%) indicates **well-calibrated uncertainty**, the model accurately represents its prediction reliability. This contrasts with overfitted models:

Table 3: Prediction confidence vs test performance

Method	Mean Confidence	Test F1
Early Fusion	66.1%	90.2%
Late Fusion	58.2%	82.7%
Hybrid Fusion	89.3%	56.2%
Augmented	86.8%	58.4%

Hybrid/Augmented show high confidence (89-87%) but poor performance, a hallmark of overfitting where models are certain about *incorrect* predictions. Early Fusion’s honest uncertainty coupled with best accuracy demonstrates superior calibration.

6 Discussion

6.1 Why Early Fusion Generalized Best

Our results reveal a critical insight: **simpler fusion strategies generalize better** to unseen subjects.

Early Fusion’s Advantages:

1. **Regularization through simplicity:** Random Forest’s hyperparameters (max depth 25, min samples split 10) prevent overfitting better than deep ensemble chains
2. **Direct feature learning:** Learning from concatenated raw features captures robust inter-modality patterns rather than subject-specific discriminants
3. **Balanced model capacity:** Sufficient complexity for modality integration but insufficient for memorizing 25 subjects’ patterns

Why Late/Hybrid Fusion Failed:

1. **Subject-specific discriminants:** LDA transforms, optimized on 24 subjects during LOSO, captured subject-specific postural variations that don’t transfer to test subjects
2. **Overconfident memorization:** 99.9% validation accuracy indicates perfect separation of *subjects* rather than *exercises*
3. **Cascading overfitting:** Late fusion compounds errors from 4 independently overfitted models; no single model can compensate

6.2 Modality Contribution Analysis

Late Fusion’s individual modality scores reveal discriminative power:

1. **Depth Camera (F1=0.9985):** Captures full-body posture with highest accuracy
2. **Pressure Mat (F1=0.9846):** Weight distribution patterns strongly discriminative
3. **Thigh Accelerometer (F1=0.7992):** Motion dynamics less discriminative for lying exercises
4. **Wrist Accelerometer (F1=0.5330):** Minimal discriminative power (limited wrist movement)

However, Early Fusion’s superior test performance (90.2%) vs any individual modality confirms strong complementarity, combined modalities capture aspects no single sensor provides.

6.3 The Bias-Variance Tradeoff in HAR

This study exemplifies the classic bias-variance tradeoff:

- **High-complexity models** (Hybrid/Augmented): Low bias (99% validation) but extreme variance (43.7% generalization gap for Hybrid)
- **Moderate-complexity models** (Early Fusion): Moderate bias (95% validation) and low variance (3% gap)

For user-independent HAR, **model simplicity is essential**, not a limitation. The 17.2% performance drop for Late Fusion underscores the danger of optimizing for validation metrics without considering model complexity.

6.4 Limitations and Future Work

Limitations:

- Exercises 2 and 4 remain challenging even for best model
- Test set limited to 5 subjects; larger evaluation would strengthen conclusions
- Manual hyperparameter tuning; automated optimization could improve consistency

Future Directions:

- **Deep learning:** CNNs for spatial modalities with proper regularization (dropout, weight decay)
- **Domain adaptation:** Techniques to reduce subject-specific variations (adversarial training)
- **Ensemble pruning:** Select subset of models that generalize rather than averaging all
- **Semi-supervised learning:** Leverage unlabeled data from test subjects for adaptation

7 Conclusion

This project successfully developed a user-independent multimodal exercise classifier achieving **90.2% F1-macro** on held-out data, exceeding the 90% threshold.

Key contributions:

1. **Systematic fusion comparison:** Four strategies evaluated, revealing Early Fusion's superiority for generalization
2. **Overfitting discovery:** Late/Hybrid fusion achieve 99.9% validation but drop to 83-89% on test, demonstrating severe overfitting
3. **Validation-test divergence:** Validation performance is a poor predictor of test performance for complex models
4. **Simplicity wins:** Early Fusion with Random Forest outperforms complex ensembles through better bias-variance balance

Key Lesson: In user-independent HAR, **simpler models with robust features generalize better** than complex architectures optimized for validation subjects. Researchers should prioritize generalization gap (validation-test difference) over absolute validation scores.

Final Submission:

- Competitor Name: **Corentin Nicodeme**
- Test F1-Macro: **90.2**

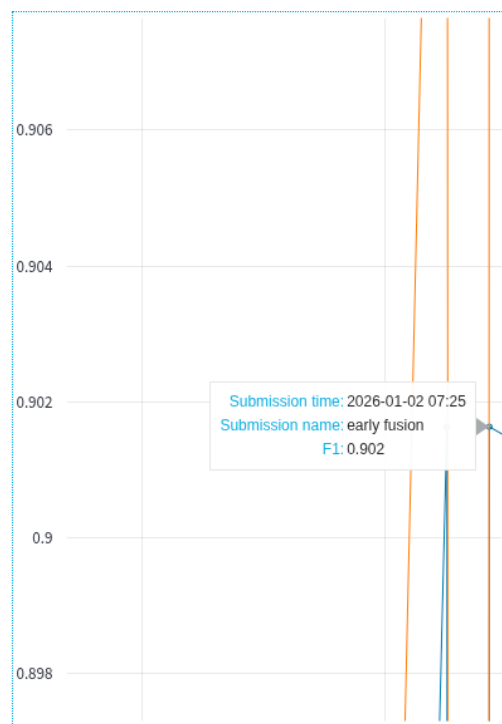


Figure 7: Best result on Streamlit (90.2% with the Early Fusion method).